

2026 9 7 DOCX XLSX Python
2026 9 7 World Bank API 7 OLS/HC3 “ ” 3
Word 3 logGDP–MMR logGDP–CPR –logGDP JSON SHA-256
/ / Word 3,615 “ 2,800 ”

		WDI 167 / SHA-256 JSON URL
Excel		“ ”D9:D11
		7
Word		“r=.489”
		BMJ NIH WHO PDF /
		/

B --- Word

XLSX “ ” D9:D11 DOCX 3 Results 6 Discussion	D9=0.5752 D10=0.4887 D11=0.4457 Word logGDP– MMR r=–.850 logGDP–CPR r=.575 –logGDP r=.489	D9 logGDP–MMR = –0.715991 n=167 D10 logGDP–CPR = 0.575205 n=139 D11 –logGDP = 0.488684 n=166 LibreOffice	GDP MMR –0.850	Word
XLSX “ ” A8:J8 DOCX Methods/Ref [3]	“1210053.data.json + 1210053.metadata.json” SHA-256 API URL	HTML V-Dem v16 2026-08-05 2026-03- 17 HTML /api/v1/indicators/... JSON 404	“ ”	JSON V- / ISO Dem
DOCX 1	Jolie [Lastname] [Institution Name] [student email]	1	/	

③ 11–16	“* *.xlsx” requirements.txt NumPy/openpyxl	CLI --input requirements.txt/
XLSX “ ” CPR -2018	28 CPR R IF	data_only
DOCX Methods	“ ” →	/
DOCX Discussion “ ”	logGDP r=.488684 VIF=1/(1-r²)=1.314 VIF=4.17 VIF	“ ” “ ” /
DOCX	3,615 URL “ 2800 ” 8 7 –8	2800 800 DOI/
XLSX /	PDF 63	/ URL /

□□□□□□□□

- XML
- Word OOXML 8
- “ ” PASS /
- “universe” 167 sample
- Flo company-reported / commercial forecast

□□□□□□□□□□□□□□□□

“ ” MMR GDP CPR CPR WDI API page=1 pages=1 per_page=20000 MMR
GDP CPR SHA-256 MMR 167/167 GDP 167/167 CPR 139/139 28 CPR
2010–2023

/ISO3	167 167 ISO3 0=27 1=51 2=55 3=33 =1 PSE
	MMR 1–993 GDP PPP 1,033,009–130,296.844 CPR 6.900–85.475 CPR 2010–2023 logGDP log(MMR) 167 LN()
MMR	n=167 116.940 SD168.263 41 Q1=9.5 Q3=155 IQR=145.5 1–993
CPR	n=139 49.896 SD20.768 54.6 Q1=31.045 Q3=66.076 IQR=35.031 6.9–85.475
CPR	n=138 PSE CPR 2015–2023 n=109
CPR	28/167 “ 0 ” 3/27=11.1% 1 4/51=7.8% 2 10/55=18.2% 3 11/33=33.3%
CPR	2010 Q1=2015 2018 Q3=2021 IQR=6 2023

□□□□□□□□□□

	n	r		Word	
–log(MMR)	166	-0.441397	-0.441397	-0.441	
–MMR	166	-0.367959	-0.367959		
–CPR	138	0.445719	0.445719	0.446	
logGDP–log(MMR)	167	-0.849733	-0.715991	-0.850	Word
logGDP–MMR	167	-0.715991	0.575205		
logGDP–CPR	139	0.575205	0.488684	0.575	Word
–logGDP	166	0.488684	0.445719	0.489	Word

Word “D9:D11” Word -0.850 0.575 0.489 logGDP–log(MMR) logGDP–CPR –
logGDP D9:D11 Word Word r D9:D11

□□□□□□□□□□□□□□□□

X=[1, predictors] OLS $s^2(X'X)^{-1}$ HC3 $(X'X)^{-1}X'\text{diag}[e_i^2/(1-h_{ii})^2]X(X'X)^{-1}$ p 95%CI t “
HC3 ”495 1e-8

□□	n / k / df	□□□□□□	R ² / □□ R ²
M1	166 / 3 / 163	β=–0.053; OLS SE=0.079, p=0.5038; HC3 SE=0.075, p=0.4812; HC3 CI [–0.202, 0.095]	0.731 / 0.728
M2	166 / 3 / 163	β=–3.487; OLS SE=10.619, p=0.743; HC3 SE=9.245, p=0.7065; HC3 CI [–21.744, 14.769]	0.520 / 0.514
M3	138 / 3 / 135	β=4.827; OLS SE=1.677, p=0.00465; HC3 SE=1.982, p=0.01616; HC3 CI [0.908, 8.747]	0.373 / 0.363
M4	138 / 4 / 134	β=4.800; OLS SE=1.687, p=0.005124; HC3 SE=2.030, p=0.01947; HC3 CI [0.786, 8.815]	0.373 / 0.359
M5	109 / 3 / 106	β=3.265; OLS SE=2.015, p=0.1081; HC3 SE=2.362, p=0.1697; HC3 CI [–1.417, 7.947]	0.342 / 0.330

M6	166 / 5 / 161	d1 β =0.103, HC3 p=0.6317 d2 β =-0.046, HC3 p=0.8232 d3 β =-0.119, HC3 p=0.6219	0.732 / 0.726
M7	138 / 5 / 133	d1 β =3.390, HC3 p=0.4939 d2 β =10.613, HC3 p=0.0403 d3 β =12.477, HC3 p=0.05269	0.376 / 0.357

M1/M2 “ ” M3 M4 HC3 p<.05 M5 109 HC3 CI 0 “ ” M7
2 0 HC3 p=.040 3 HC3 p=.053 CI 0 “ / ”

Python

	Python 3.9.6 NumPy openpyxl
	HC3 df t p CI p “<0.0001” 0
/	5–171 C/E/H/K/L M6/M7 code=0
	WDI/OWID 167 SHA “ → ”
	glob /

[1] WHO		2000 339/2020 223 PDF HTML PDF
[2] World Bank WDI		API 2026-07-13 lastupdated 2026-09-03 “ ”
[3] V-Dem/OWID		V-Dem(2026) Democracy report v16 2026-08-05 JSON/
[4] Mozilla		2022-08-17 18/25 10 period+10 pregnancy apps+5 wearables Mozilla warning
[5] Grundy et al., BMJ		BMJ 2019;364:1920 DOI BMJ 403 19/24 FemTech
[6] FTC Flo		2021-06-22 Facebook/Google FTC “FTC alleged/complaint”
[7] FTC Premom		2023-05-17 proposed FTC order 100,000 proposed
[8] Flo		2024-10-01 2022 3
[9]		150 2024 39.29bn 2030 97.25bn CAGR16.3/16.37%
[10] Nature Rev Bioeng		Editorial 2024-10-11 2:797–798 DOI 5% 2021 2.5bn/ 3%
[11] Nat Biomed Eng		Editorial 2026-07-17 10:1265–1266 DOI
[12] NIH ORWH		PDF 403 10.8% 4,466m /
[13] WHO Endometriosis		2025-10-15 10%/190m
[14] WHO PCOS		10–13% reproductive-aged women WHO
[15] Flo		2024-06-19 380m downloads 67m 70m Word “nearly70m” [16]
[16] Flo		2024-07-30 “as of June 2024 nearly70m MAUs”

Mozilla 18/25 BMJ 19/24 Flo FTC Premom FTC

- Excel “ ”D9=-0.715991 D10=0.575205 D11=0.488684 data_only
- ISO3→ [3] “ ”
- Word 1 PDF

- requirements.txt Python
- 3,615

□□□□□

- [5] [12] /DOI [8] “company-reported”
- 63
- “ HC3 ” p
- VIF=1.314 4.17

□□□□□□□□□□□□

9 7 DOCX XLSX audit_artifacts/ WDI CSV JSON/ 2026
403/ API “ ” “ ”